

LESSON 8-1 • HOMEWORK PACKET • SHEET 1 OF 2**Equations and Solutions**

Maryland Standard 6.EE.B.5 • Learning target – I can check whether a value is a solution of an equation by substituting it.

REMEMBER

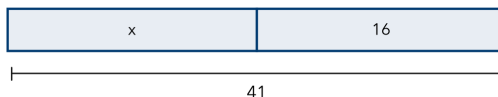
- A **solution** is a value of the variable that makes an equation **true**.
- To check a value, **substitute** it for the variable. Then compare the two sides.
- If the two sides are not equal, the value is **not** a solution.

WORKED EXAMPLEIs $n = 14$ a solution of $n + 9 = 23$?**Step 1** Replace n with 14: $14 + 9 = 23$ **Step 2** Compare: $23 = 23$. Both sides match.**Answer Yes, $n = 14$ is a solution.****PRACTICE***show your work • label units*

1. Substitute the value. Write
- T**
- for true or
- F**
- for false.

- a. $y + 8 = 20$ ($y = 12$) ____ b. $5m = 35$ ($m = 6$) ____ c. $k - 9 = 16$ ($k = 25$) ____
- d. $h \div 4 = 7$ ($h = 32$) ____ e. $13 + w = 30$ ($w = 17$) ____ f. $6p = 54$ ($p = 9$) ____
- g. $q - 11 = 4$ ($q = 15$) ____ h. $9r = 63$ ($r = 8$) ____

2. The tape diagram shows
- $x + 16 = 41$
- . Which value is the solution: 15, 25, or 35? Substitute each value to check.

**Solution** _____

- 3.
- Bike Rack.**
- A bike rack held
- b
- bikes. After 7 bikes left, 18 bikes were still there. Kendra writes
- $b - 7 = 18$
- . Is
- $b = 11$
- a solution? Is
- $b = 25$
- a solution? Show both checks.

*Show your work.***Solution** _____

- 4.
- Find the Error.**
- Andre says
- $p = 8$
- is a solution of
- $4p = 12$
- because
- $4 + 8 = 12$
- . Explain his mistake. Then find the real solution.

- 5.
- Stretch.**
- Malik paid \$42 for
- t
- movie tickets that cost \$6 each, so
- $6t = 42$
- . Is
- $t = 36$
- a solution? Is
- $t = 7$
- ? Explain which value makes sense.

Solution _____**CHECK YOURSELF**

- ☐
- I can do this on my own
- ☐
- I need a little help
- ☐
- I have a question about #____

AT HOME Make up an equation for someone at home. Check their guess together by substituting.

LESSON 8-1 • HOMEWORK PACKET • SHEET 2 OF 2**Writing Equations**

Maryland Standard 6.EE.B.6 • Learning target – I can write an equation for a situation and test a proposed answer.

REMEMBER

- Use a **variable** for the unknown amount.
- “7 less than n ” is $n - 7$, **not** $7 - n$.
- **Test** an answer by substituting it.

WORKED EXAMPLE b books plus 15 more is 52 books. Test $b = 37$.**Step 1** Equation: $b + 15 = 52$ **Step 2** Test: $37 + 15 = 52$, so it checks.**Answer** $b + 15 = 52$; $b = 37$ **PRACTICE***show your work • label units*

- Write an equation for each sentence. Use n for the number.
 - A number increased by 9 is 31. _____
 - 7 times a number is 63. _____
 - 12 less than a number is 20. _____
 - A number divided by 5 is 8. _____
- Stickers.** Rosa gave away 14 stickers and has 29 left. Write an equation for s . Test $s = 43$.

**Equation and test** _____

- Field Trip.** 56 students ride equally in 4 vans. Write an equation for v , the students per van. Test $v = 12$ and $v = 14$.

*Show your work.***Equation and solution** _____

- Reading Log.** Luis reads the same number of pages, p , each day for 5 days. He reads 85 pages in all. Write an equation. Test $p = 17$.

Equation and test _____

- Find the Error.** Tia is 13. She is 4 years older than her brother. Omar writes $b - 4 = 13$ for his age. Explain the mistake. Find b .

TEST PRACTICE*choose the best answer*

- Jada saved \$9 a week for w weeks. She saved \$72. Which equation fits?

A $w + 9 = 72$
B $9w = 72$
C $w \div 9 = 72$
D $72w = 9$
- Which value of n is the solution of $n - 15 = 27$?

A $n = 12$
B $n = 27$
C $n = 42$
D $n = 405$

CHECK YOURSELF
☐ I can do this on my own
 ☐ I need a little help
 ☐ I have a question about #____